

Don't Make Me Repeat Myself

Analyzers & More For Quality by Default

Daniel Ward

Me

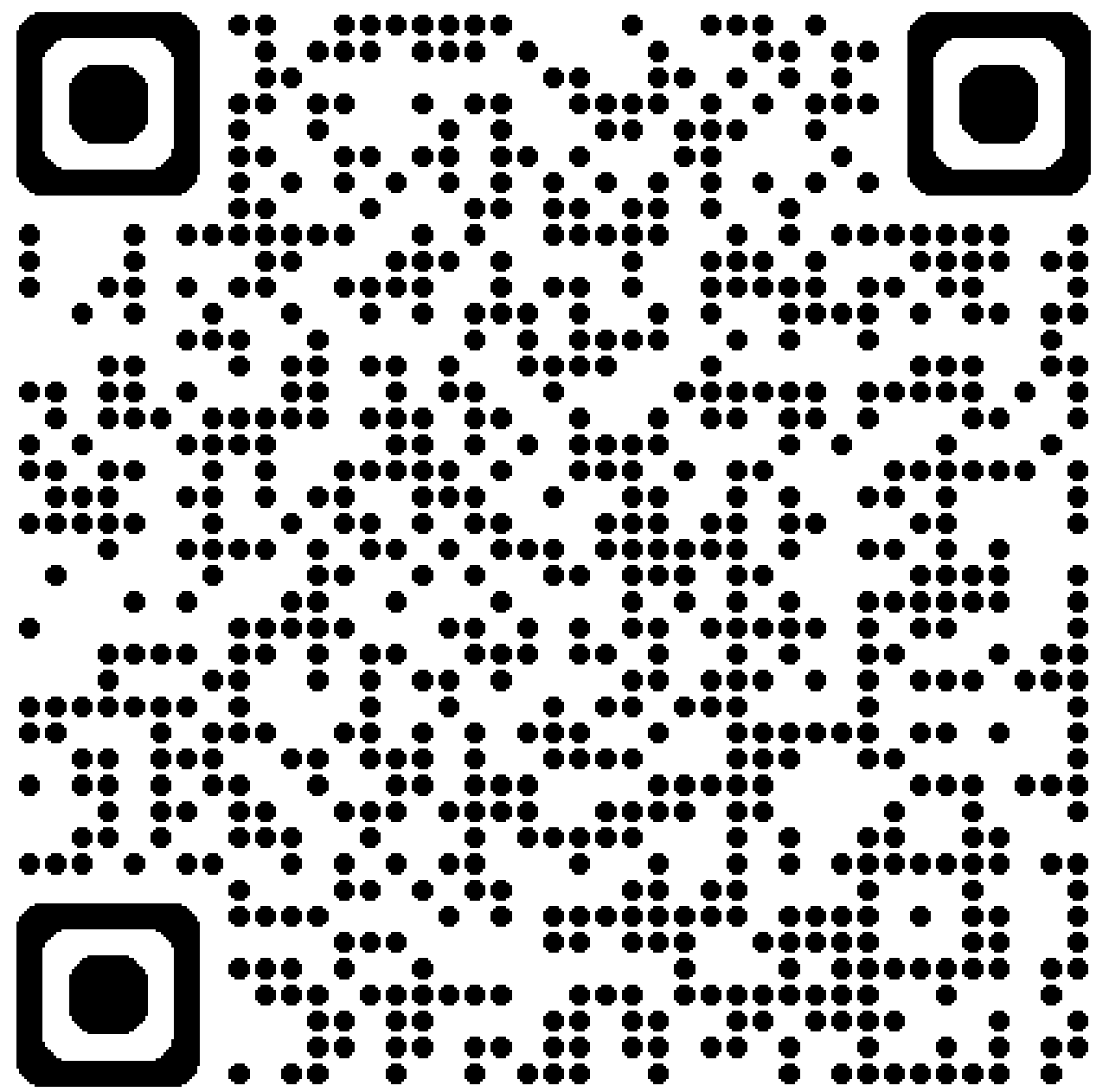


LEAN
TECHNIQUES



- Software developer, consultant
- Microsoft .NET MVP
- Co-organizer of the San Antonio/Austin .NET User Group
-  daninacan.com
-  [@danielwarddev](https://twitter.com/danielwarddev)
-  [daniel-ward-dev](https://www.linkedin.com/in/daniel-ward-dev)
-  danielwarddev.bsky.social





Outline

- The problem
- dotnet format
- Csharpier
- Roslyn analyzers
- Architecture tests

The problem



code-enjoyer commented 2 days ago

Contributor



Use new style for lists

```
- var list = new List<string>();  
+ List<string> list = [];
```



reasonable-teammate commented 2 days ago

Member



What's wrong with you? This sucks!



code-enjoyer commented 2 days ago

Contributor



I hate you! I quit!

The problem

- Disagreements over...
 - Styling
 - Formatting
 - Whitespace
 - Conventions
 - etc.

Napkin math

- 3 PRs per week
- \$50/hr salary for devs
- 15 minutes to run PR checks
- 15 minutes for develop to tweak code
- 1 tweak = 15 mins (tweak) + 15 mins (PR checks)
- $(15 + 15) * 3 = 90$ mins of dev time per week
- $\$50 * 1.5 = \textbf{\$75/week} * \textbf{52 weeks} = \textbf{\$3900/year}$ in dev time

Shifting left

Software Development Lifecycle



Shifting left – why?

- 15 mins on each PR adds up over time
- Less painful for you
- Don't have to think about it anymore, just happens
- Fewer defects in production
- Cheaper to fix mistakes
- Avoid technical debt
- Less rework

Benefits of shifting left

- Fewer defects in production
- Cheaper to fix mistakes
- Avoid technical debt
- Less rework

dotnet format

- Configurable code formatter
- Ships with .NET
- Uses an .editorconfig if present, otherwise default settings
- `dotnet format --verify-no-changes`
- Use `EnforceCodeStyleInBuild` to get analyzer warnings on build
- Combine with `TreatWarningsAsErrors`

CSharpier

- <https://csharpier.com>
- Opinionated code formatter
- dotnet tool install csharpier
- dotnet csharpier format .
- Plugin for lots of IDEs
 - Visual Studio
 - VS Code
 - Rider
 - Neovim
 - Emacs
 - Zed

CSharpier

- Can run:
 - CLI
 - Pre-commit hook
 - On build (with CSharpier.MSBuild)
 - Programmatic API
- Config: `.csharpierrc`, `.csharpierrc.json/yaml`, `.editorconfig`
- `dotnet csharpier check .`
 - Throws error if any changes

Roslyn analyzers

- You're already using these!
 - (probably)
- The Roslyn compiler is exposed as a platform to extend
- Built-in and custom

Roslynator

- <https://github.com/dotnet/roslynator>
- Prebuilt Roslyn analyzers for common use cases
- <https://josefpihrt.github.io/docs/roslynator/analyzers>
- Also has its own CLI
 - dotnet roslynator fix
- Other options like StyleCop, SonarLint, etc.

Architecture tests

- Test the architecture of your code
- If it's in code, you can enforce it
 - Naming classes/methods/etc a certain way
 - Vertical slices don't depend on each other
- Test-time, not build-time

ArchUnitNET

- <https://github.com/TNG/ArchUnitNET>
- Library for writing architecture tests in C#
- Fork of ArchUnit for Java
- Analyzes C# bytecode
- Real tests in your suite

AI Bonus

- May be able to remove some of your coding rules in AI instructions
- Deterministic feedback always preferable!

Recap

- The problem (shifting left)
- dotnet format
- CSharpier
- Roslyn analyzers
- Architecture tests

Thank you!

